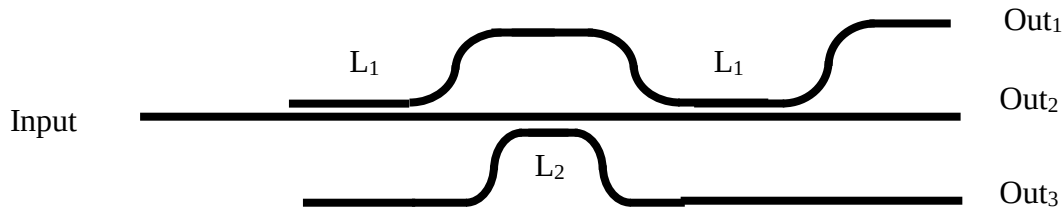


Sheet 3

Question 1:

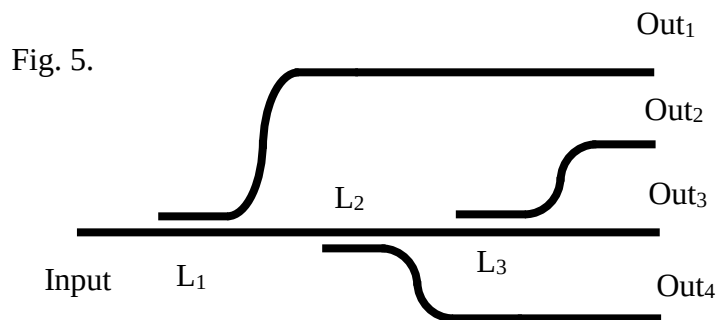
The IO circuit shown in Fig. is based on the use of two coupled waveguides as a directional coupler with a coupling length $L_c = 2$ mm at the wavelength $\lambda = 1.5$ μm . The circuit is fabricated on a glass substrate and the guiding structure has an effective refractive index of 1.5. If an optical power of 3 mW is injected at the circuit input, calculate the optical power at the outputs assuming a length difference of $\Delta L = 3$ μm due to the bending for each bend existing in the circuit and neglect the bending losses. The first and third couplers have lengths of $L_1 = 1.6$ mm while the second coupler has a length of $L_2 = 1$ mm as shown in Fig.. If the wavelength of the input is changed to 1.55 μm , calculate the three outputs assuming that the effective refractive indices are the same. \



The IO circuit composed of directional couplers.

Question 2:

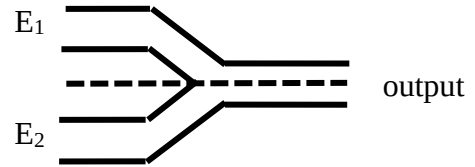
The IO circuit shown in Fig. is based on the use of two coupled waveguides as a directional coupler with a coupling length $L_c = 1$ mm at the wavelength $\lambda = 1.55$ μm . Calculate the lengths L_1 , L_2 , and L_3 of the three couplers used, such that the four outputs of the circuits have the same optical power at $\lambda = 1.55$ μm ? What will be the output powers (as a percentage of the input power) at the wavelength $\lambda = 1.3$ μm assuming that the effective refractive indices of the modes do not change significantly between these two wavelengths.



Question 3:

For the, the symmetric Y junction shown in Fig. 1, find the output when:

- a- $E_1 = E_2$
- b- $E_1 = -E_2$
- c- $E_1 = E_2/2$
- d- $E_1 = -E_2/2$

**Question 4:**

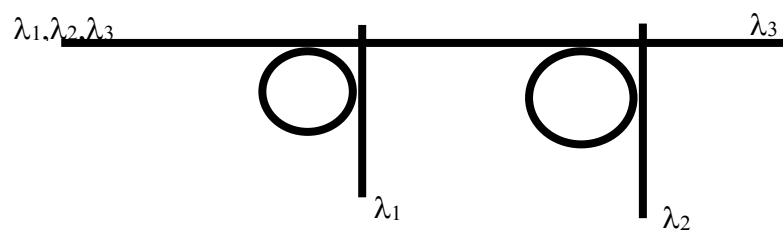
An unbalanced Mach-Zehnder interferometer with a path difference between the two arms equals $1 \mu\text{m}$ is fabricated on a glass substrate of refractive index 1.48 using the ionic exchange process that produces $\Delta n = 5 \times 10^{-3}$. The width of the waveguide is $4 \mu\text{m}$. Evaluate the transfer function of the interferometer (output/input) at the wavelengths $\lambda = 1, 1.3$ and $1.55 \mu\text{m}$. Assume that only the fundamental mode is excited in the interferometer.

Question 5:

A balanced Mach-Zehnder interferometer is fabricated on an electrooptic material. If the output of the interferometer was zero at an applied voltage of 30 Volts, what will be the output at $V = 10, 20$ and 25 volts. Assume that the generated phase shift is linearly proportional to the applied voltage.

Question 6:

Design an optical ring resonator triplexer that enables to separate the 1300 nm, 1490 nm and 1550 nm using two cascaded rings as shown in Figure. In your design determine the radius of each ring, the waveguide width and refractive index and assume a certain coupling coefficient between the ring and the straight guide.

**Question 7:**

A ring resonator is fabricated on a Si waveguide with refractive index $n = 3.5$ surrounded by air ($n = 1$). The width of the waveguide is $0.2 \mu\text{m}$ and the diameter of the ring is $30 \mu\text{m}$. If the coupling coefficient between the ring and the adjacent waveguides is 10%, calculate:

- a- The free spectral range of the resonator
- b- The finesse of the resonator.